

Meeting 6 Supplemental — Technical Work Done

Satellite Images for Emissions Tracking · Since Meeting 5 (April 10-22, 2026) · Prepared by Vincent Cornelius

1. Multi-Year Historical Backfill Across All 8 European Sites

At Meeting 5, quantification results existed only for two Netherlands sites (Groningen and Maasvlakte) from single acquisitions. The pipeline has now been extended to produce a structured, multi-year time series across all eight target sites, covering 2019 through 2025.

Sites covered:

- Weisweiler (Germany) — lignite plant
- Rybnik (Poland) — coal complex
- Bełchatów (Poland) — Europe's largest coal plant
- Lippendorf (Germany) — lignite plant
- Neurath (Germany) — lignite plant
- Boxberg (Germany) — lignite plant (clean control)
- Groningen (Netherlands) — gas field
- Maasvlakte (Netherlands) — gas terminal

Each acquisition runs the full pipeline: Sentinel-2 ingestion → CH4Net v8 inference → signal/control detection → CEMF+IME emission rate estimation with ERA5 wind. Results are written to a structured JSON time series (`results_analysis/historical_backfill_timeseries.json`) that can be consumed directly for downstream analysis. Full per-site, per-year detections are in Appendix B (summary) and Appendix C (5-year S/C grid).

Key structural findings, all traceable to the JSON record for the cited acquisition:

- Bełchatów is the cleanest repeated signal: $S/C = 849.08$ on 2020-06-01, with CFAR detections in 2020, 2021, and 2024 ($S/C = 142.98$).
- Lippendorf shows a strong single-date signal in 2024 ($S/C = 315.98$ on 2024-06-26) but no repeated detection across the 2020-2023 window.
- Rybnik peaks at $S/C = 309.55$ on 2023-06-01 but does not trigger CFAR on any of its 5 valid acquisitions (high terrain variability).
- Boxberg's 2024-07-21 acquisition shows $S/C = 1202.83$ — this is almost certainly a terrain/cloud artifact on the control annulus, not a methane plume, and is consistent with the 'clean control' designation (no CFAR detection across any acquisition). Flagged for manual audit under the backfill integrity lockdown.
- Groningen shows one CFAR detection (2024-08-17, $S/C = 6.01$) across 6 acquisitions.
- Maasvlakte and Weisweiler have max S/C values far below their Meeting-5 numbers because ERA5 + CEMF is now wired in — earlier point estimates relied on climatological wind fallbacks that inflated inferred flow. See Appendix A.

2. Data-Quality Fix — Partial-Swath Tile Degeneracy

During the backfill run, a systematic failure mode was identified and resolved. Certain Sentinel-2 orbital passes (R051 and R108 tracks) produce 'partial-swath' tiles where the plant location falls entirely outside the actual image swath boundary. The satellite tasking system still generates a tile record, but 60–85% of pixels in that tile are zero. CH4Net inference was still being run on these blank tiles, producing a statistically degenerate probability map — a uniform signal that made the signal/control ratio equal exactly 1.0, which looked like a valid 'no detection' rather than a missing observation.

The fix involved three steps:

- Identified the fingerprint: any record where $\text{site_mean} \approx \text{ctrl_mean}$ and $\text{sc_ratio} = 1.0$ exactly.
- Confirmed root cause by measuring the valid-pixel fraction in the 100×100-pixel crop centred on each plant location.
- Repaired 7 affected records (5× Weisweiler, 1× Bełchatów, 1× Neurath) by reclassifying them as 'no_coverage' and nulling the SC-derived fields.
- Added a coverage pre-check to the pipeline that catches this before inference runs on future acquisitions.

This matters directly for quantification: any site where the valid-pixel fraction is below 50% will now be correctly excluded from flow rate estimation rather than silently producing a zero-emission record.

3. Statistically Rigorous Detection Threshold (Conformal Calibration)

Previously the detection threshold was a fixed heuristic (signal/control ratio $\geq 1.15 + 3\sigma$). This has been replaced with a threshold calibrated using split conformal prediction, a method with a *provable* false-positive rate guarantee rather than an approximation.

Method. Held-out non-emitter sites were sampled from CORINE Land Cover across Europe (industrial, agricultural, and grassland terrain), Sentinel-2 tiles were downloaded and run through CH4Net, and the empirical distribution of signal/control ratios at known-clean sites was used to set the threshold at the $\alpha = 0.10$ quantile. This guarantees $\Pr(\text{false positive}) \leq 10\%$ with a finite-sample bound, not asymptotically.

Results on $n = 14$ non-emitter calibration sites (full table in Appendix D):

- $\tau = 4.1052$ at $\alpha = 0.10$ (production threshold, FPR $\leq 10\%$ guarantee; empirical FPR on calibration set = 0.0%)
- $\tau = 2.5653$ at $\alpha = 0.20$ (more sensitive threshold, FPR $\leq 20\%$; empirical FPR = 14.29%)
- Legacy threshold (1.15) for comparison: empirical FPR = 78.57% on the same $n = 14$ calibration set — i.e., 11 of 14 verified non-emitters would fire as false positives
- Bootstrap 90% CI on τ ($n = 2000$ resamples): $[2.5653, 4.1052]$, mean 3.5846 ± 0.7237

This directly answers one of the EIB's open questions from Meeting 5 about calibration against regulatory thresholds. The detection threshold now has an auditable statistical basis that can be stated to a regulator or model risk reviewer. Mondrian stratification per ecoregion has also been computed (Appendix E) but with small-n warnings on 4 of 5 ecoregions — this is the case for expanding the calibration set to 30+ sites.

4. Quantification Pipeline Scaled to 8 Sites (CEMF + IME + ERA5)

The CEMF+IME+ERA5 quantification layer, introduced at Meeting 5 on the two Netherlands sites, has now been scaled into the backfill loop. For every acquisition where S/C exceeds the calibrated threshold, the pipeline automatically executes CEMF spectral retrieval, IME inversion, and live ERA5 wind retrieval, and writes the result alongside the detection record with full provenance: flow rate Q in kg/h, uncertainty interval, annualized tonnes, IRA Waste Emissions Charge at the 2026 statutory rate (\$1,500/tonne), wind speed, wind source (ERA5 vs. climatological fallback), CEMF validity flag, and scene quality classification.

ERA5 integration remains the single most consequential upgrade relative to the pre-Meeting 5 pipeline. The 3.5 m/s climatological fallback it replaces was the dominant source of systematic error in the emission rate estimate, and each output record now logs exactly which wind source was used. Six quantification records have been produced so far — 4 included, 2 governance-excluded — spanning Groningen, Maasvlakte, Bełchatów (two dates), Lippendorf, and Neurath. Full per-record flow rates, wind provenance, plume pixel counts, exclusion flags, and 2026 IRA liability numbers are in Appendix A.

5. Strategic Reorientation: From Detector to MRV / Model-Validation System

A project-wide reframing has been developed and documented in the CH4Net Strategic Directions v2 memo. The core pivot: stop treating CH4Net as a detector to make better; treat it as a live model being independently validated, with full uncertainty decomposition at every step. Practically this means every number carries a decomposed, traceable uncertainty; every model decision has a known false-positive rate and detection floor per operating regime; and every domain transfer (Central Asian O&G; → European coal) is accompanied by an explicit out-of-distribution score. The five prioritized workstreams map cleanly onto the SR 11-7 / SS1/23 three-pillar structure.

Work stream	Focus	Status
WS1	End-to-end quantification (conformal calibration, uncertainty decomposition, ensembles)	In progress — conformal threshold done; decomposition script built; heteroscedastic CEMF and CQR pending
WS2	Foundation-model fine-tuning + out-of-distribution detection (Prithvi-EO-2.0 via LoRA)	Planned — addresses domain shift (Central Asia → EU) with a provable OOD score

WS3	Physics-informed advection-diffusion surrogate (FNO + SBI) as conceptual-soundness anchor	Planned
WS4	Historical backfill with active learning (TypiClust + Margin) for efficient EU labeling	Backfill pipeline built and running; active-learning layer planned
WS5	Detection threshold calibration, synthetic plume injection, stress testing	Conformal threshold calibrated (done, Appendix D-E); synthetic plume injection pending

6. New and Modified Scripts

The following scripts have been added or significantly changed since Meeting 5. Pull from the main branch before running anything. A public GitHub repository with one-command conda setup and Makefile-orchestrated reproduction is in active preparation.

Script	Status	Purpose
scripts/historical_backfill_timeseries.py	NEW	Runs the full pipeline (ingest → inference → detection → CEMF/IME) across all 8 sites, 2019–2025. Outputs historical_backfill_timeseries.json.
scripts/repair_backfill_coverage.py	NEW	One-shot repair tool for partial-swath degeneracy. Checks valid-pixel fraction in site crop; reclassifies affected records as no_coverage. Already applied — 7 records fixed.
scripts/sample_nonemitter_sites.py	NEW	Samples verified non-emitter sites from CORINE Land Cover for the conformal calibration set.
scripts/download_nonemitter_tiles.py	NEW	Downloads Sentinel-2 L1C tiles for the sampled non-emitter sites.
scripts/run_nonemitter_inference.py	NEW	Runs CH4Net inference on non-emitter tiles to build the calibration distribution.
scripts/calibrate_conformal_threshold.py	NEW	Applies split conformal prediction to calibrate the S/C detection threshold. Outputs τ at user-specified α levels (default 0.10 and 0.20).
scripts/uncertainty_decomp.py	NEW	Decomposes flow rate uncertainty into four sources: wind (ERA5 spread), CEMF coefficient, plume mask, background annulus. Produces per-detection breakdown.
src/quantification/cemf.py	EXISTING	CEMF spectral retrieval — converts CH4Net probability map to total plume mass in kg.

src/quantification/ime.py	EXISTING	IME inversion — converts plume mass to flow rate Q (kg/h) with uncertainty bounds.
src/quantification/era5_utils.py	EXISTING	ERA5 wind client — queries Copernicus CDS API for 10m U/V components at any lat/lon/time.

Appendix A — Quantification Workstream Technical Note

The quantification pipeline closes a specific gap: CH4Net identifies where a methane plume sits and maps the affected pixels, but a pixel map has no units. A financial risk model cannot use it. Workstream 2 converts CH4Net binary detection masks into emission rate estimates in kg/h, with annualised tonnes and IRA Waste Emissions Charge liability values at the 2026 statutory rate of \$1,500/tonne.

Methodology (five sequential steps)

- **Spectral retrieval (CEMF, Varon et al. 2021).** Sentinel-2 B11 ($\approx 1,610$ nm) serves as a spectrally adjacent reference; B12 ($\approx 2,190$ nm) is $\sim 5\times$ more methane-sensitive. Per-pixel reflectance anomalies are projected onto the methane absorption signature using a sensitivity of 4×10^{-7} reflectance per ppb·m, integrated across plume pixels, and converted to total plume mass in kg.
- **ERA5 wind retrieval.** Copernicus CDS supplies hourly 10m U/V at 0.25° resolution, matched to acquisition time and plume centroid. This replaces the 3.5 m/s climatological fallback that was the dominant error source. Every record logs whether wind came from ERA5 or the fallback.
- **IME inversion.** $Q = (M \times U) / L$ converts instantaneous plume mass to kg/h. L is estimated from the bounding geometry of the detection mask; uncertainty bounds of $\pm 40\%$ follow the literature value for single-overpass multispectral retrievals.
- **Annualisation and liability.** Annual tonnes assume continuous emission (upper-bound). The IRA Waste Emissions Charge at \$1,500/tonne is applied as a financial proxy, since it is the only enacted statutory methane liability rate globally as of 2026. EU Methane Regulation / CBAM calibration is a planned extension.
- **Structured JSONL output.** One record per detection: scene ID, acquisition timestamp, centroid, Q with bounds, wind speed and source, CH4Net peak probability, cloud-cover class, CEMF validity flag, and retrieval notes. Streamable, appendable, and feeds directly into downstream climate scenario models.

All quantification records produced to date (n = 6)

Full contents of results_analysis/quantification.json. Every number here is the raw output of the pipeline, not summary statistics. Rows flagged EXCLUDED are kept in the record for provenance but are not rolled up into site-level aggregates; the exclusion reason is logged on each record.

Site	Wind	Source	Q (kg/h)	CI	Annual t	IRA 2026	Pixels	Status
Groningen	5.5 m/s	ERA5	28.8	17.3–40.3	252	\$377,906	1,980	EXCLUDED (CFAR FP)
Maasvlakte	4.5 m/s	ERA5	426.5	255.9–597.1	3,736	\$5,604,210	1,565	included

Bełchatów	3.5 m/s	fallback*	475.9	285.6–66.3	4,169	\$6,253,589	56,374	included (σ -inflated)
Lippendorf	3.5 m/s	fallback*	1,579.6	947.8–2,211.5	13,838	\$20,756,338	219,207	EXCLUDED (terrain)
Neurath (2024-06-25)	3.5 m/s	fallback*	534.7	267.4–802.1	4,684	— (σ -inflated)	14,614	included, $\pm 50\%$
Bełchatów (2024-08-24)	2.19 m/s	ERA5	426.4	298.5–554.4	3,735	— (EU ETS TBD)	5,714	included, $\pm 30\%$

* 'fallback' = ERA5 unavailable at acquisition time; the pipeline used the 3.5 m/s climatological prior and flagged the record with WIND_FALLBACK, which triggers the σ -inflation governance rule (+30 percentage points on wind σ , propagated through the Monte Carlo uncertainty budget).

Groningen is CFAR-suppressed as a likely false positive (S/C below the conformal threshold); Lippendorf is excluded pending terrain-artifact audit (the 2024-06-26 acquisition sits on the calibration frontier). These exclusions are what the backfill-integrity lockdown in Next Target #1 is designed to systematise.

Uncertainty budget — why $\pm 40\%$ is a placeholder

The reported $\pm 40\%$ bound is inherited from Varon et al. (2021) and is not decomposed, not calibrated to this project's data, and not defensible under technical scrutiny. A rough decomposition identifies three contributors: wind-speed uncertainty ($\sim 15\text{--}20\%$), plume-length geometric estimation ($\sim 15\text{--}20\%$), and CEMF spectral retrieval ($\sim 10\text{--}15\%$). In quadrature these give $\approx \pm 28\text{--}30\%$ for ERA5-integrated retrievals over quasi-homogeneous surfaces. Replacing the placeholder with a per-scene, source-attributed budget (σ_{wind} , σ_{CEMF} , σ_{mask} , $\sigma_{\text{background}} \rightarrow$ Monte Carlo combination) is the WS1 priority and the script is already built. Wrapping the result in a conformalized quantile regression interval calibrated on Sherwin hold-out data is the bundled sprint goal.

Reference

Varon, D.J., Jervis, D., McKeever, J., Spence, I., Gains, D., and Jacob, D.J. (2021). High-frequency monitoring of anomalous methane point sources with multispectral Sentinel-2 satellite observations. *Atmospheric Measurement Techniques*, 14, 2771–2785. <https://amt.copernicus.org/articles/14/2771/2021/>

Appendix B — Per-Site Backfill Summary (2019-2025)

Rolled up from `historical_backfill_timeseries.json`. 'N records' counts all attempted acquisitions including failures; 'N valid' excludes partial-swath and no-coverage records. 'Max S/C' is the single-acquisition maximum on valid records. CFAR count is the number of acquisitions that triggered the legacy detection rule ($\tau \geq 1.15 + 3\sigma$); conformal detection ($\tau = 4.1052$) is stricter and is not yet computed per-record.

Site	N records	N valid	Max S/C	Best date	CFAR detections	Note
Weisweiler	11	6	23.46	2024-09-18	1	RWE; 5 partial-swath repaired
Rybnik	5	5	309.55	2023-06-01	0	PGE; high σ_{ctrl} suppresses CFAR
Bełchatów	8	7	849.08	2020-06-01	3	PGE; strongest repeated signal
Lippendorf	8	8	315.98	2024-06-26	1	LEAG/EnBW; single-date spike
Neurath	8	7	67.20	2024-08-29	2	RWE; TROPOMI-confirmed 2024-06-25
Boxberg	8	7	1202.83	2024-07-21	0	LEAG; clean control, 2024-07-21 flagged for audit
Groningen	6	6	6.01	2024-08-17	2	NAM (Shell/Exxon); CFAR detections suppressed by conformal τ
Maasvlakte	7	7	0.67	2024-06-28	0	Uniper/Engie; no backfill detection

Totals across the 8 sites: **61 acquisitions ingested, 53 valid** after the partial-swath repair, **9 CFAR detections** on the legacy threshold. Applying the conformal threshold $\tau = 4.1052$ reduces this to the subset that survives a provable FPR $\leq 10\%$ guarantee — the count is dominated by Bełchatów (3 acquisitions clear τ : 849.08, 18.53, 142.98) with Neurath, Lippendorf, and Weisweiler each contributing one above- τ acquisition.

Appendix C — Per-Site × Per-Year S/C Grid

Maximum signal/control ratio on the best valid acquisition for each site-year. Entries marked with * triggered CFAR detection (legacy threshold $1.15 + 3\sigma$). '—' indicates no valid acquisition in that year (typically due to partial-swath or cloud cover).

Site	2020	2021	2022	2023	2024
Weisweiler	0.14	2.82*	—	—	23.46
Rybnik	39.97	0.99	0.31	309.55	2.01

Bełchatów	849.08*	18.53*	0.54	0.18	142.98*
Lippendorf	1.15	1.23	0.15	0.90	315.98
Neurath	0.51	0.60	0.71	0.43	67.20*
Boxberg	0.97	0.03	0.00	1.14	1202.83
Groningen	4.60	0.25	0.35	4.65	6.01*
Maasvlakte	0.00	0.25	0.01	0.00	0.67

Reading the grid: the Bełchatów column is the load-bearing signal for the EU coal backfill — three of five years clear legacy CFAR and two of those three (849.08 in 2020, 142.98 in 2024) clear the conformal $\tau = 4.1052$ threshold. Every other site has at most one above- τ acquisition. Boxberg's 1202.83 in 2024 is a terrain artifact candidate (clean control, no CFAR trigger) and is first on the integrity-lockdown queue.

Appendix D — Conformal Calibration Set (n = 14 Non-Emitters)

Full contents of results_analysis/nonemitter_sc_scores.json. These are the held-out non-emitter acquisitions used to calibrate the conformal detection threshold. Sites were sampled from CORINE Land Cover across 5 European ecoregions with a minimum 50 km separation from any known emitter. sc_cfar is the CFAR-normalised score that feeds into the conformal quantile; sc_ratio is the raw signal/control ratio for comparison.

ID	Location	Ecoregion	CLC class	sc_ratio	sc_cfar
001	Lüneburg Heath, DE	Atlantic	pasture	1.14	1.2465
002	Belgian Ardennes, BE	Atlantic	mixed_forest	1.21	1.3212
003	Moselle Valley, DE	Continental	arable_land	84.57	4.1052
004	Hamburg rural, DE	Atlantic	arable_land	1.00	1.3331
005	Šumava NP, CZ	Continental	coniferous_forest	143.90	2.1172
007	Bohemian Forest, CZ	Continental	coniferous_forest	0.83	1.3547
008	Franconian Highland, DE	Continental	complex_cultivation	753.97	2.7397
010	NE Slovakia lowlands	Pannonian	arable_land	0.92	1.1302
011	Łódź region, PL	Continental	arable_land	2.19	1.8626
013	Mazovian lowland, PL	Continental	broadleaved_forest	2.08	2.5653
014	Schleswig-Holstein, DE	Atlantic	pasture	1.00	0.9984

0 1 5	Swedish coastal plain	Boreal	mixed_forest	0.80	1.3317
0 1 6	Baltic coast, LV	Boreal	coastal_wetland	0.77	1.2984
0 1 7	Apennines, IT	Mediterranean	mixed_forest	0.002	0.0020

Sorted `sc_cfar` distribution: [0.002, 0.998, 1.130, 1.247, 1.298, 1.321, 1.332, 1.333, 1.355, 1.863, 2.117, 2.565, 2.740, 4.105]. At $\alpha = 0.10$ the conformal quantile lands on the maximum ($\tau = 4.1052$), which is *conservative by construction* for $n = 14$ — the calibration set is small enough that the threshold is single-observation-dominated (nonemit_003, Moselle Valley, arable land with very low `ctrl_mean` inflating the raw S/C). This is the primary motivation for expanding to $n \geq 30$ with proper ecoregion stratification (Next Target #1, Meeting 6 agenda).

Appendix E — Mondrian Per-Ecoregion Calibration

Stratified conformal thresholds computed per ecoregion from the same $n = 14$ calibration set. Mondrian stratification gives per-stratum FPR guarantees but loses statistical power rapidly when within-stratum sample sizes drop; 4 of 5 ecoregions currently carry a small- n warning and their thresholds should be treated as indicative rather than binding until the set is expanded.

Ecoregion	n	Sites in stratum (<code>sc_cfar</code>)	τ ($\alpha=0.10$)	Legacy FPR ($\tau=1.15$)
Atlantic	4	0.998, 1.247, 1.321, 1.333	1.3331	75.0%
Continental	6	1.355, 1.863, 2.117, 2.565, 2.740, 4.105	4.1052	100.0%
Pannonian	1	1.130	1.1302	0.0%
Boreal	2	1.298, 1.332	1.3317	100.0%
Mediterranean	1	0.002	0.0020	0.0%

The Continental stratum is the only one with a non-degenerate sample size ($n = 6$), and its $\tau = 4.1052$ matches the global threshold. Atlantic ($n = 4$, $\tau = 1.33$) is noticeably more permissive but still far above the legacy 1.15 heuristic, which fires on 3 of 4 Atlantic non-emitters at the 75% legacy FPR. For the Meeting 6 backfill sites: Bełchatów and Rybnik sit in Continental; Neurath, Lippendorf, Weisweiler, Boxberg in Continental; Groningen and Maasvlakte in Atlantic. Applied per-ecoregion, the Atlantic sites face a lower bar to clear — which is one reason Groningen's S/C = 6.01 is interesting even though it would fail the global τ .